

Bronze Layer — Raw Ingestion

ServiceTrack Project | Execution Screenshots

Screenshot 1: Customer Data Schema and Row Count

The screenshot shows a Jupyter Notebook interface with a Python kernel. The code in the notebook counts the total number of records in a DataFrame named `df_customers_raw`. The output section displays the schema of the DataFrame and a sample of the data.

```
7 # Count and print the total number of records
8 customers_count = df_customers_raw.count()
9 print(f"Total customer records: {customers_count}")
```

Output

```
root
 |-- customer_id: string (nullable = true)
 |-- customer_name: string (nullable = true)
 |-- phone_number: long (nullable = true)
 |-- email: string (nullable = true)
 |-- city: string (nullable = true)
 |-- registration_date: date (nullable = true)
```

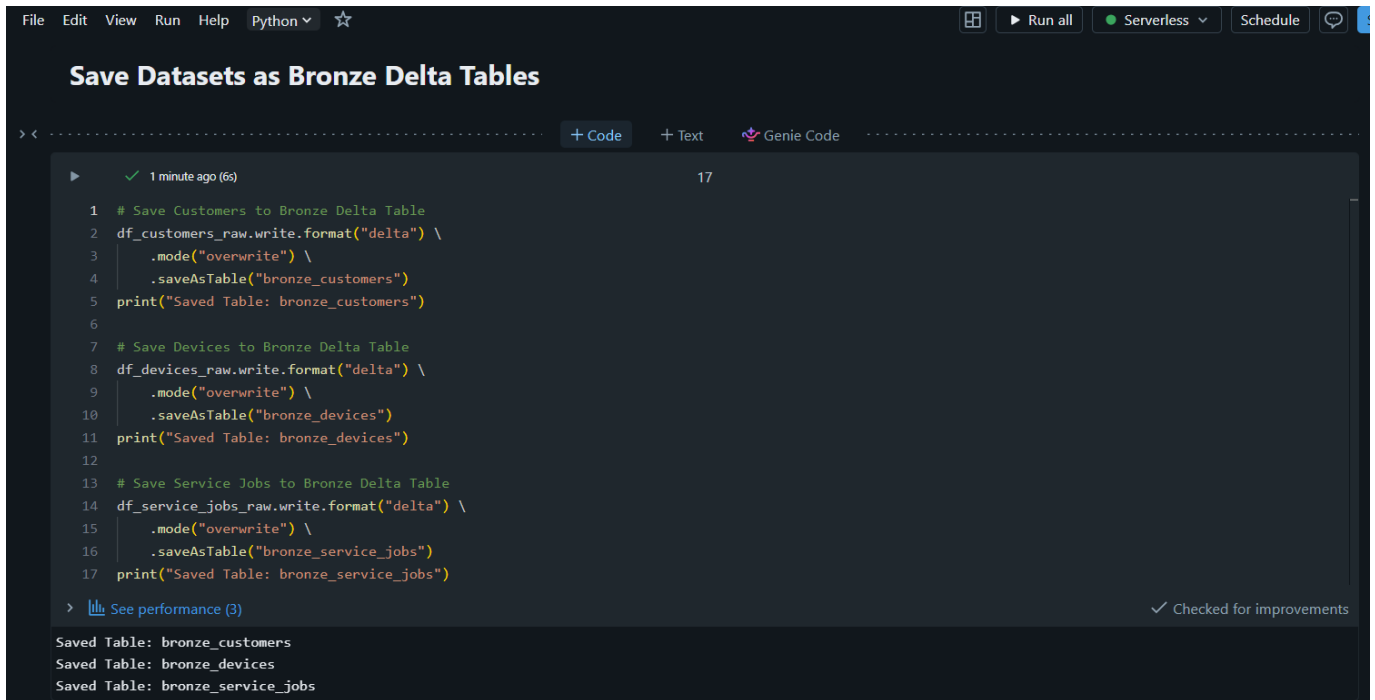
	customer_id	customer_name	phone_number	email	city	registration_date
1	CUST0001	Kiran Patel	9433218196	kiran.patel5@gmail.com	Delhi	2022-08-12
2	CUST0002	Pallavi Patel	9838637940	pallavi.patel98@yahoo.com	Nagpur	2023-03-09
3	CUST0003	Bhavna Tiwari	9235116155	bhavna.tiwari78@outlook.co...	Indore	2022-02-14

5 rows | 5.706s runtime

Total customer records: 300

Raw customers.csv loaded into a PySpark DataFrame — schema, sample records, and total row count (300) verified.

Screenshot 2: Save Datasets as Bronze Delta Tables



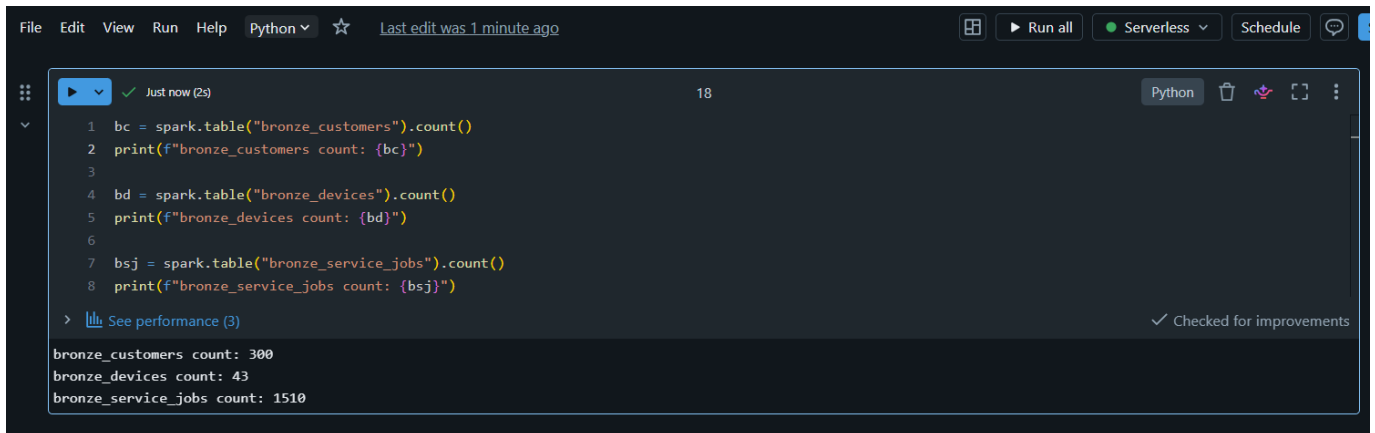
The screenshot shows a Jupyter Notebook interface with a dark theme. The title bar at the top includes 'File', 'Edit', 'View', 'Run', 'Help', 'Python', and a star icon. On the right, there are buttons for 'Run all', 'Serverless', 'Schedule', and a chat icon. The notebook title is 'Save Datasets as Bronze Delta Tables'. Below the title bar, there are tabs for '+ Code', '+ Text', and 'Genie Code'. The code cell is titled '1 minute ago (6s)' and contains 17 lines of Python code. The code defines three functions to save raw datasets as Bronze Delta Tables: 'Save Customers to Bronze Delta Table', 'Save Devices to Bronze Delta Table', and 'Save Service Jobs to Bronze Delta Table'. Each function uses 'df_raw.write.format("delta")' to write the data in Delta format, followed by '.mode("overwrite")' and '.saveAsTable()' to save it as a Bronze Delta Table. The output of the code cell shows three lines: 'Saved Table: bronze_customers', 'Saved Table: bronze_devices', and 'Saved Table: bronze_service_jobs'. At the bottom right of the code cell, there is a link 'See performance (3)' and a status 'Checked for improvements'.

```
1 # Save Customers to Bronze Delta Table
2 df_customers_raw.write.format("delta") \
3     .mode("overwrite") \
4     .saveAsTable("bronze_customers")
5 print("Saved Table: bronze_customers")
6
7 # Save Devices to Bronze Delta Table
8 df_devices_raw.write.format("delta") \
9     .mode("overwrite") \
10    .saveAsTable("bronze_devices")
11 print("Saved Table: bronze_devices")
12
13 # Save Service Jobs to Bronze Delta Table
14 df_service_jobs_raw.write.format("delta") \
15     .mode("overwrite") \
16     .saveAsTable("bronze_service_jobs")
17 print("Saved Table: bronze_service_jobs")
```

Saved Table: bronze_customers
Saved Table: bronze_devices
Saved Table: bronze_service_jobs

All three raw datasets (customers, devices, service jobs) written to Delta format and saved as Bronze tables.

Screenshot 3: Bronze Table Row Count Verification



The screenshot displays a Jupyter Notebook interface with a dark theme. At the top, there is a menu bar with 'File', 'Edit', 'View', 'Run', 'Help', and a 'Python' dropdown. To the right of the menu bar are icons for 'Run all', 'Serverless', 'Schedule', and a chat icon. Below the menu bar, the notebook content is shown. It includes a code cell with 8 lines of Python code that uses the Spark API to count rows in three tables: 'bronze_customers', 'bronze_devices', and 'bronze_service_jobs'. The code is as follows:

```
1 bc = spark.table("bronze_customers").count()
2 print(f"bronze_customers count: {bc}")
3
4 bd = spark.table("bronze_devices").count()
5 print(f"bronze_devices count: {bd}")
6
7 bsj = spark.table("bronze_service_jobs").count()
8 print(f"bronze_service_jobs count: {bsj}")
```

Below the code cell, the output is displayed, showing the row counts for each table:

```
bronze_customers count: 300
bronze_devices count: 43
bronze_service_jobs count: 1510
```

At the bottom right of the code cell, there is a status bar that says 'Checked for improvements'.

Row counts for all three Bronze Delta tables confirmed: bronze_customers = 300, bronze_devices = 43, bronze_service_jobs = 1510.